



Review Article

Integrating human biomonitoring into food safety risk assessment: Advances, challenges, and future perspectives

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ABSTRACT

Human biomonitoring (HBM) is a vital tool for evaluating internal exposure to chemicals and their metabolites. It provides a direct reflection of the cumulative body burden from various sources and pathways. Although extensive data has been generated through numerous projects, linking these findings to health risks remains challenging. This review emphasises the importance of HBM in risk assessment and explains how to interpret biomonitoring data. Through concrete examples, including national case studies from Singapore and China and chemical-specific assessments of pesticides and mixtures, we demonstrate the practical application of HBM in food safety. The strengths and limitations of the HBM approach, as well as its future directions, are discussed in detail.

1. Introduction

With the development of agricultural production systems, ongoing advances in food processing technologies, and the increasing integration of global food trade, chemical contamination of food has become a global and persistent challenge with important implications for the safety and integrity of the food supply chain (Rather et al., 2017). Chemical contaminants in food originate from multiple sources across the entire “farm to fork” continuum, encompassing environmental media, agricultural inputs, food processing operations, and food contact materials (Li, 2025). Major sources of contamination include: (1) environmental background contamination, such as heavy metals (e.g. lead and mercury) and persistent organic pollutants (Wang et al., 2023); (2) residues arising from the use of agricultural inputs, including pesticides (Fu et al., 2023); (3) contaminants formed during food processing, such as acrylamide (da Costa et al., 2023); (4) chemical substances migrating from food contact materials (Tanaviyutpakdee and Karnpanit, 2023); (5) naturally occurring toxic substances in food, including phytotoxins and mycotoxins (Awuchi et al., 2021); and (6) deliberate illegal adulteration motivated by economic gain (Zhang and Xue, 2016). The health risks associated with chemical contaminants in food are complex and depend on the intrinsic hazardous properties of the substances, their sources and forms, as well as the route, level, and duration of exposure.

Given the diversity of contamination sources and exposure

pathways, the accurate characterisation of human exposure represents a key challenge in chemical health risk assessment. Traditional exposure assessment is typically based on combining measured concentrations of chemicals in food with food consumption data to estimate external exposure at the individual level. However, this approach relies heavily on limited monitoring data and a series of simplifying assumptions, making it sensitive to shortcomings in sample representativeness, parameter extrapolation, and model specification, which can compromise accurate characterisation of exposure variability across populations and individuals (Kettler et al., 2015). In particular, under conditions involving multiple food sources, heterogeneous consumption patterns, and long-term low-level exposure, estimates based on external exposure often fail to adequately reflect the actual internal chemical burden (Authority, 2009).

In this context, human biomonitoring (HBM) has emerged as a complementary approach for exposure assessment. HBM is commonly defined as the assessment of internal exposure to chemicals by measuring the levels of these substances, their metabolites, or related biomarkers in biological matrices, such as blood, urine, hair, teeth, saliva, breast milk, and nails (Choi et al., 2015). The key advantage of HBM is its ability to provide a comprehensive measure of total human exposure, which integrates chemicals from all routes and sources, including air, diet, and personal care products. This approach provides a more accurate assessment of total body burden and serves as a

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(Ougier et al., 2021a). Building on these foundations, PARC was initiated to support European and national authorities in chemical risk assessment and risk management and to advance the regulatory application of HBM. Within PARC, 17 interlinked HBM projects address key methodological and application-oriented challenges, including the development of a sustainable European HBM framework, improved data reusability, the establishment of HBM guidance values (HBM-GVs), the enhancement of chemical mixture risk assessment, and the linkage of chemical exposures with health outcomes. In addition, the European Food Safety Authority (EFSA) typically performs dietary exposure assessments by combining food consumption data from specific population groups with chemical concentration databases. HBM data provide complementary evidence to this approach by directly informing on internal exposure. For instance, EFSA has systematically integrated serum biomonitoring data on per- and polyfluoroalkyl substances (PFAS), showing that perfluorooctanoic acid (PFOA), perfluorononanoic acid (PFNA), perfluorohexane sulfonic acid (PFHxS), and perfluorooctane sulfonic acid (PFOS) are the main contributors to body burdens and that infants and children represent particularly highly exposed groups. By compiling HBM data across different time periods, EFSA assessed temporal trends in PFAS exposure. Furthermore, EFSA combined internal dose information with physiologically based pharmacokinetic (PBPK)

Major HBM initiatives include the Human Biomonitoring for Europe (HBM4EU, 2016–2022), the Partnership for the Assessment of Risks from Chemicals (PARC, 2022–2029), the National Health and Nutrition Examination Survey (NHANES, 1999–present), the Canadian Health Measures Survey (CHMS, 2007–present), and the China National Human Biomonitoring programme (CNHBM, 2017–present) (Cao et al., 2021; Apel et al., 2020; Snapkow et al., 2024; Haines et al., 2017; Ahluwalia et al., 2016). Among these, HBM4EU represented a major advance in establishing a harmonised and quality-assured HBM framework, encompassing the entire workflow from study design and sampling strategies to the analysis and interpretation of biomonitoring data

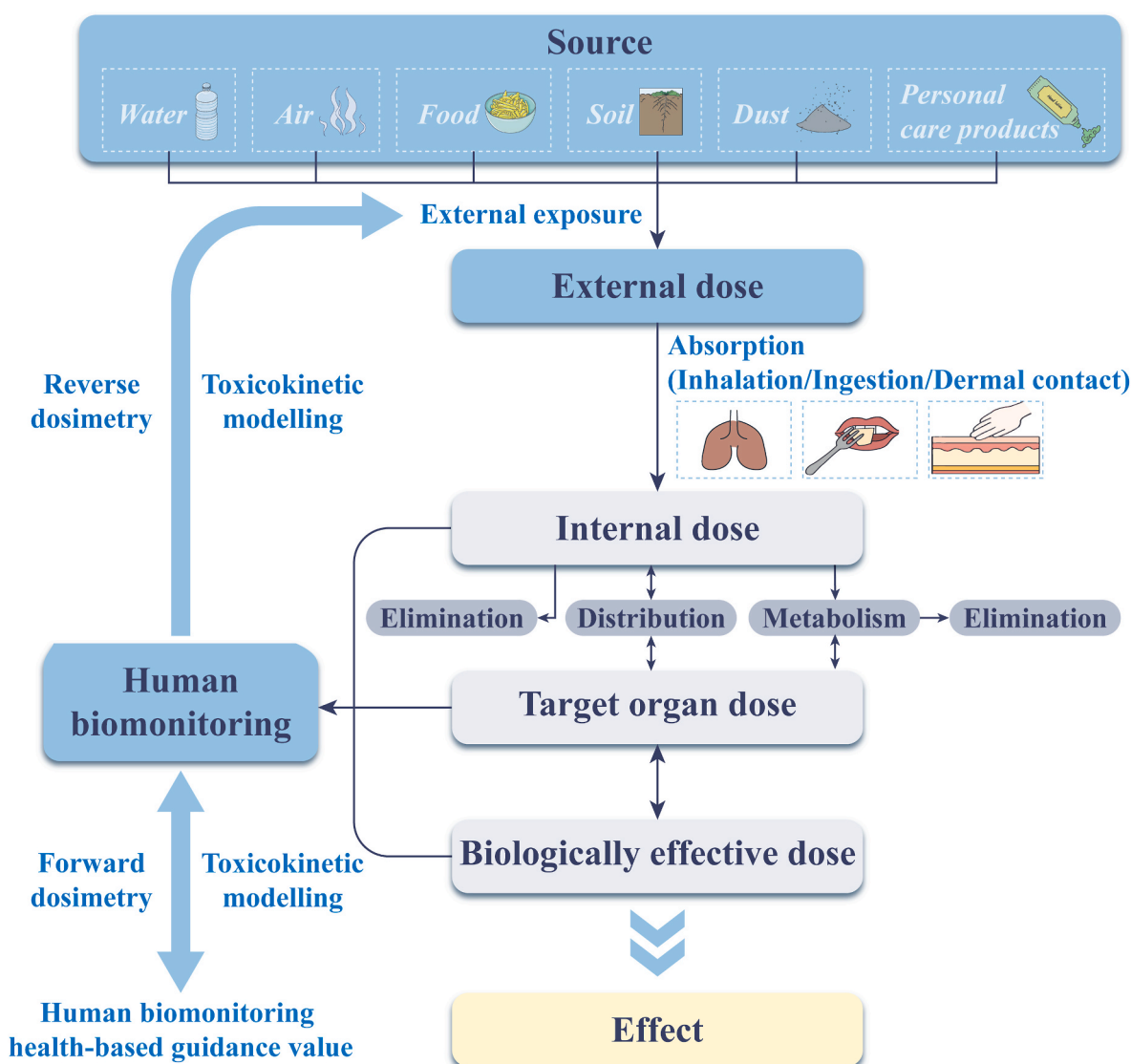


Fig. 1. Role of human biomonitoring in risk assessment.

modelling to support the derivation of a tolerable weekly intake for PFAS mixtures and to inform population risk characterisation (Schrenk et al., 2020). This work illustrates the joint use of traditional dietary exposure assessment and HBM evidence in food safety risk assessment. Moreover, the integrated approach exemplified here for PFAS is methodologically applicable to other food-related chemicals, such as bisphenol A (BPA).

Given the growing complexity of global environmental pollution and the continuous emergence of new chemicals, traditional risk assessment methods may not accurately reflect actual exposure levels and associated health risks. In contrast, HBM provides internal exposure data and serves as a valuable complement to traditional assessment methods. Therefore, this review examines how HBM can be applied across the risk assessment framework. By addressing current challenges and exploring future directions, the review aims to establish a robust theoretical and practical foundation for the use of HBM in ensuring food safety.

2. Materials and methods

2.1. Identifying relevant studies

This study employed a narrative literature review to provide an overview of the role of HBM in health risk assessment. Relevant studies were identified through structured database searches supplemented by targeted manual searches. Embase, PubMed, and Web of Science were searched, together with the official websites of HBM4EU and PARC, the EFSA Supporting Publications, and the EFSA Journal. The search covered publications from 1 January 2010 to 31 December 2025 with no language restrictions and focused on original research articles.

2.2. Study selection

All records identified were imported into EndNote 21 for reference management and duplicate removal. Studies were screened using pre-defined inclusion and exclusion criteria. Eligible studies were original research articles published between 1 January 2010 and 31 December 2025 that examined the application of HBM in health risk assessment, with a focus on food safety.

Studies were excluded if they met any of the following criteria: (1) conference abstracts or proceedings, preprints, errata, or similar publication types; (2) review articles; (3) studies primarily focused on animals or plants rather than humans; (4) studies addressing only health effects or disease outcomes without exposure assessment; (5) studies not relevant to the objectives of this review; or (6) studies primarily addressing occupational exposure, including workers in food production or processing industries. The overall study framework is shown in Fig. 2.

3. The role of human biomonitoring and biomarkers in food safety risk assessment

3.1. Integration of human biomonitoring into risk assessment

According to the WHO/IPCS framework, the risk assessment process begins with problem definition and comprises four essential stages: hazard identification, hazard characterization, exposure assessment, and risk characterization (Meek et al., 2011). In this context, HBM is a valuable tool that provides data on actual substance levels in the body, supporting each stage of the risk assessment process.

Although HBM is not the main tool for identifying toxicity mechanisms in the hazard identification stage, it can provide evidence of exposure levels in the human body by comparing the concentrations of

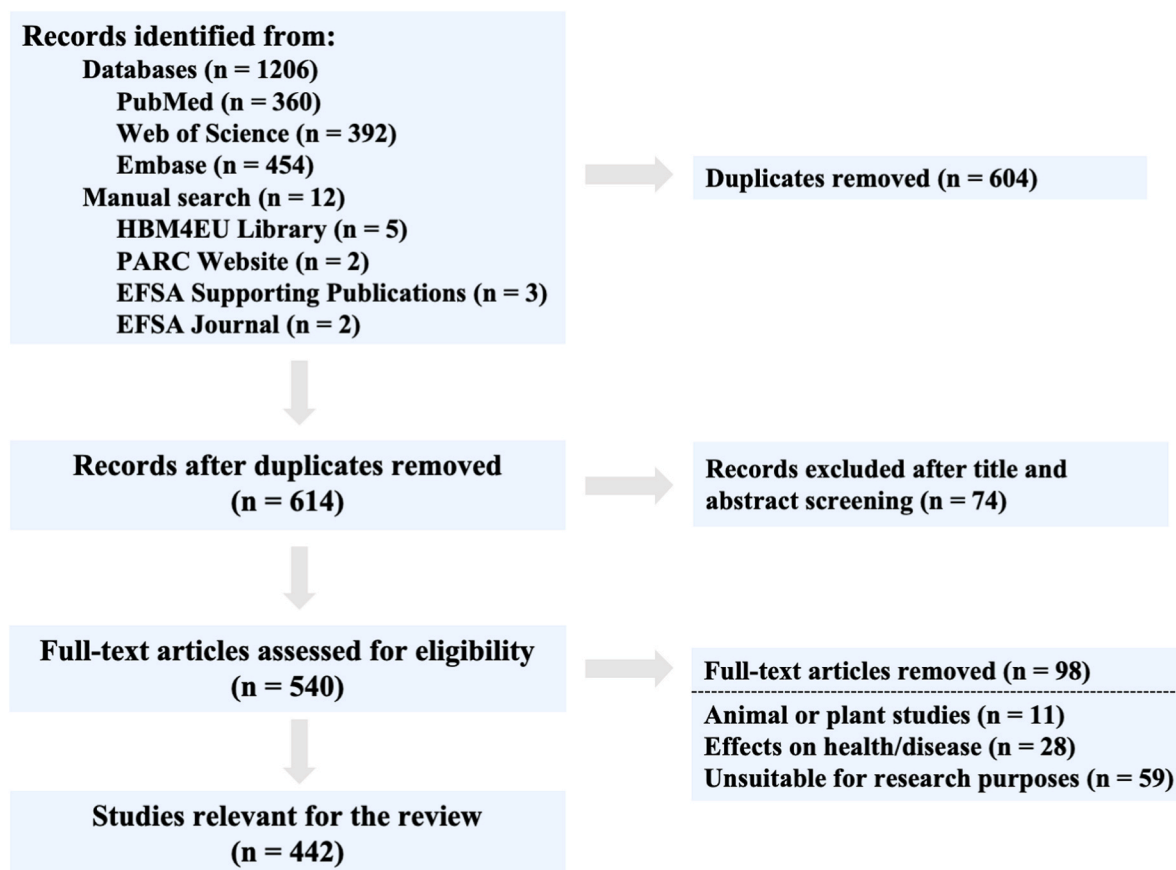


Fig. 2. The flow chart depicting the study selection process for this review.

chemicals and their metabolites in exposed and control groups (Manno et al., 2010). This can help to validate or support potential toxicity hypotheses raised in animal and in vitro studies. For instance, the EFSA employed HBM data to analyze the toxicokinetics of BPA in humans during its re-evaluation. At the same time, HBM data provided crucial evidence to explore potential links between BPA exposure and its effects on metabolism, the nervous system, reproduction, development and the immune system (Lambré et al., 2023). Such applications depend on the availability of high-quality and reliable biomonitoring data, and the robustness of national HBM data for risk assessment is therefore underpinned by rigorous quality assurance protocols. For example, CNHBM has established comprehensive quality control strategies for the targeted quantitative analysis of chemical contaminants to ensure the accuracy and comparability of its biomonitoring data (Yang et al., 2025). In the hazard characterisation stage, long-term HBM data, together with epidemiological and toxicological evidence, can support the evaluation of internal dose–response relationships (Bevan et al., 2017). HBM primarily reflects internal dose through biomarker concentrations of chemicals or their metabolites, whereas assessment of biological responses requires studies with appropriate health endpoints and ideally longitudinal follow-up.

When conducting exposure assessments, HBM integrates data from various sources to provide an accurate reflection of the body's overall chemical burden. Although internal exposure levels in target organs or tissues are considered the “gold standard” for assessment, they are often difficult to measure directly in population studies. Therefore, HBM, using minimally invasive samples such as blood, or non-invasive samples such as urine and hair, is regarded as the most practical and effective method of assessing chemical exposure (Zare Jeddi et al., 2022). In risk characterization, when environmental monitoring data is unreliable, missing or difficult to measure accurately, the internal dose measured by HBM can be used directly to assess the actual risk to individuals or populations (Bevan et al., 2017). By measuring internal doses and health effect biomarkers, HBM can reveal differences in individual susceptibility caused by genetic variations or physiological conditions. This makes it possible to identify high-risk groups. When certain groups are found to have significantly higher exposure levels than others, this provides governments with crucial evidence to develop targeted actions. For example, since the Canadian government identified di(2-ethylhexyl) phthalate (DEHP) as a substance posing potential health risks in 1994, it has introduced several regulatory measures to reduce its impact, including listing it on the *Cosmetic Ingredient Hotlist*, enforcing the *Phthalates Regulations*, and establishing provisions under the *Food and Drugs Act* (Karthikeyan et al., 2024).

3.2. Application of biomarkers for assessing dietary exposure

HBM relies on the measurement of biomarkers, broadly defined as measurable indicators of exposure, effect, or susceptibility (Hernández-Jerez et al., 2024). In recent years, biomarkers have also been increasingly applied in nutrition and diet-related research to support the assessment of food intake and dietary patterns, and are widely regarded as a complementary approach to traditional questionnaire-based dietary assessment methods. Initiatives such as the Food Biomarkers Alliance, established under the European Joint Programming Initiative “A Healthy Diet for a Healthy Life”, have promoted the systematic identification and validation of biomarkers associated with the intake of specific foods and food groups (Wishart, 2024). Although these dietary intake biomarkers are not biomarkers of exposure to chemical contaminants, they remain relevant to dietary exposure assessment because they can support the characterisation of food consumption patterns that may contribute to chemical exposure. Examples of such biomarkers, together with related methodological information, are summarised in [Supplementary Table S1](#).

4. Interpretation of human biomonitoring data

Although HBM provides direct evidence of internal exposure, biomarker measurements alone are insufficient for health risk assessment. The key challenge is to translate HBM data into information relevant for regulatory and scientific decision-making. Accordingly, HBM data can be interpreted using qualitative or quantitative approaches (Poddalgoda et al., 2024).

4.1. Qualitative approaches

The qualitative approaches assess population exposure by integrating biomonitoring data with information on toxicokinetics and use patterns of chemicals. For example, Health Canada concluded that beryllium, trichlorooxo vanadium, and vanadium oxide pose a low risk to human health based on multiple lines of evidence. These included very low or non-detectable concentrations in large-scale biomonitoring studies when evaluated against analytical detection and quantification limits, toxicokinetic evidence of rapid elimination and limited bio-accumulation, as well as restricted industrial use without consumer applications, suggesting limited population-wide exposure (H.C. Environment and Climate Change Canada, 2020).

4.2. Quantitative approaches

4.2.1. Direct comparison

In the direct comparison method, biomarker concentrations measured in HBM studies are directly compared with internal levels known to be associated with adverse health effects. This approach requires robust evidence establishing quantitative relationships between biomarker concentrations and health outcomes, derived from epidemiological studies, human volunteer trials, or animal experiments, as exemplified by Health Canada's assessments of PFOS and lead (Canada, 2006, 2013). Owing to these stringent data requirements, direct comparison is used less frequently than forward or reverse dosimetry and remains applicable to only a limited number of substances. For example, plastic-related chemicals are widely detected in HBM studies but generally lack well-defined internal concentration–health effect relationships required for direct comparison.

4.2.2. Forward dosimetry

Within the forward dosimetry framework, HBM values serve as key tools for the health-based interpretation of biomonitoring data. According to the German Human Biomonitoring Commission, these values are classified into HBM-I and HBM-II. HBM-I represents an internal concentration below which no adverse health effects are expected, whereas HBM-II serves as an action level above which health risks increase and immediate risk management measures are warranted. Concentrations between HBM-I and HBM-II indicate a range of potential concern in which adverse effects cannot be excluded (Angerer et al., 2011).

Biomonitoring equivalents (BEs) constitute another important category of reference values. BEs represent concentrations of a chemical or its metabolites in biological media that correspond to established external exposure guidance values. Their derivation relies on different methodological approaches, including PBPK modelling, as demonstrated for metals such as aluminium and titanium and for organic chemicals such as BPA (Ougier et al., 2021b; Poddalgoda et al., 2021a; Poet et al., 2010), correlations between biomarker concentrations and external intake, as shown for molybdenum (Hays et al., 2016), and mass-balance approaches commonly applied to derive urinary BEs for substances such as barium, iodine, and tin (Poddalgoda et al., 2017; Murray and D'Amour, 2016). In practice, BE values serve as screening benchmarks, with concentrations well below the BE indicating low overall exposure, whereas values approaching or exceeding the BE warrant further investigation of exposure sources.

When suitable external exposure guidance values are lacking, biomonitoring guidance values (BGVs) have been developed as an alternative framework for interpreting HBM data. BGVs directly link biomarker concentrations to critical biological effects using PBPK modelling, as illustrated by the derivation of a BGV for chlorpyrifos based on cholinesterase inhibition (Arnold et al., 2015). At the European level, the HBM4EU initiative has established a harmonised framework for deriving HBM-GVs. The HBM-GV is defined as the concentration of a substance or its metabolites in human biological material below which adverse health effects are not expected over a lifetime of exposure.

From conceptual and practical perspectives, these reference values serve distinct roles. HBM-I and HBM-II are action-oriented values based on the assumption of effect thresholds and are primarily applied to support national-level risk management decisions. BEs function as conversion and screening tools that relate HBM data to existing external exposure benchmarks, with interpretation dependent on the toxicokinetic models and parameters applied. HBM-GVs provide a harmonised, tiered, and evidence-based framework at the European level that explicitly addresses uncertainty and supports coordinated interpretation of HBM data across countries.

Within PARC, the first set of HBM-GVs and corresponding substance dossiers, covering selected pyrethroid insecticides, plasticizers, metals, and neonicotinoid insecticides, is being finalised (PARC, 2022). In parallel, international initiatives such as the publicly accessible Human Biomonitoring Health-Based Guidance Value Dashboard support harmonised interpretation of HBM data by compiling guidance values from multiple sources (Nakayama et al., 2023; Group, 2022).

Table 1 summarises food-related chemical substances, their corresponding biomarkers and analytical matrices, major food sources, and available HBM health-based guidance values.

4.2.3. Reverse dosimetry

Reverse dosimetry is a method that uses biomonitoring data to estimate external exposure levels through reverse modelling. Various methods have been developed to reconstruct exposure from HBM data, which can be broadly classified into deterministic and stochastic approaches.

Deterministic methods include mass-balance approaches and physiologically based toxicokinetic (PBTK) methods (Gurusankar et al., 2017). The mass-balance approach is the simplest method for estimating daily intake, particularly for compounds that are rapidly metabolized and eliminated within 24 h. Using the urinary mass-balance equation, Health Canada derived daily intake estimates from HBM data for the following phthalates: butyl benzyl phthalate (BBP), dibutyl phthalate (DBP), and DEHP (Jeong et al., 2020; Zidek et al., 2017). Additionally, the PBTK model-based approach is applicable to a wide range of compounds, including those that are rapidly metabolized and those with slower metabolic clearance. Kwon et al. (2024) used data on blood lead levels from the Korea National Environmental Health Survey to estimate lead exposure levels through a PBTK model.

Conversely, stochastic methods rely on more complex computational methodologies, including the exposure conversion factor (ECF) and Bayesian methods. The ECF approach generally assumes steady-state conditions and has been applied in draft exposure assessments for boron (S.C.I.P.A.C.J.J.o.t. Oil and C.C.A. group, 2025). The Bayesian approach can handle complex, dynamic exposure scenarios and manage the various uncertainties and variabilities introduced during modelling. It is often used alongside Markov chain Monte Carlo (MCMC) algorithms and PBTK models for exposure reconstruction. For instance, McNally et al. (2012) employed a PBTK model, Bayesian inference and MCMC simulation to reconstruct the inhalation exposure levels of the volatile organic compound m-xylene from HBM data.

4.2.4. Integration of forward and reverse dosimetry

Although both forward and reverse dosimetry rely on pharmacokinetic modelling to establish a quantitative link between internal doses

and external exposure, the concepts and applications of forward and reverse dosimetry in HBM differ significantly.

Forward dosimetry typically begins with an established external exposure guidance value, using PBTK models to predict internal doses. In recent years, generic PBTK models have been increasingly explored as tools to link external exposure and internal dose, particularly when substance-specific kinetic data are limited. Pletz et al. (2020) assessed the performance of two generic models, IndusChemFate (ICF) and the High-Throughput Toxicokinetics (HTTK) package. The ICF model can simulate metabolic processes and predict metabolite concentrations, but uncertainties in input parameters and limited scope for parameter refinement led to substantial deviations from measured HBM data, including underestimation of urinary concentrations for phthalates. By contrast, the HTTK model enables population-based simulations based on pre-defined physiological and physicochemical parameters, but does not explicitly simulate metabolism and tended to underestimate internal concentrations for phenols and phthalates, largely due to uncertainty in clearance estimates. When considering individual pharmacokinetic variations, the variability in predicted internal concentrations remains relatively controllable (Hays et al., 2007). Therefore, forward dosimetry is particularly appropriate for chemicals with long half-lives that tend to accumulate in the body, as it enables the establishment of reference values that reflect safe thresholds for chronic exposure. However, while external guidance values have been established for many chemicals, corresponding HBM health-based guidance values remain limited, and any limitations or biases in the external guidance values may be reflected in the derived HBM health-based guidance values (Becker et al., 2012). Furthermore, BE values are generally derived based on the assumption of long-term, steady-state exposure. When exposure includes short-term peaks, such as the consumption of large amounts of high-selenium seafood over a brief period, BEs may fail to capture the risk associated with transiently high internal doses (Turck et al., 2023). Additionally, the use of non-specific biomarkers reduces the reliability of BEs. For example, whole-blood selenium reflects only total selenium concentration. Distinguishing between the toxicological effects of organic and inorganic forms would require more specific biomarkers, significantly increasing technical complexity (Combs, 2015).

In reverse dosimetry, biomarker concentrations are used alongside toxicokinetic models to estimate external exposure. This method is particularly useful for chemicals with complex exposure sources, such as methylmercury. Nevertheless, a key limitation of reverse dosimetry is that it is non-unique by essence, meaning that different exposure scenarios may result in the same internal concentration (Sohn et al., 2004). Consequently, doses derived from HBM data tend to fall within a wide potential range. The reliability of dose estimates depends on the accuracy of the parameters and an in-depth understanding of potential exposure scenarios. When relevant data are lacking, or the parameters used do not reflect the characteristics of the target population, dose estimates may be inaccurate. For instance, directly extrapolating fractional urinary excretion data obtained from adults to infants and young children may result in an under- or overestimation of exposure (Stroli Benedetti et al., 2005). A linear regression analysis has been used to demonstrate the correlation between arsenic concentrations in drinking water and arsenic levels in hair (Agusa et al., 2014). However, this model only explained around 40% of the variability, suggesting that individual toxicokinetic differences and other uncontrolled factors could significantly impact the reliability of the estimates.

Although forward and reverse dosimetry have distinct advantages, limitations and applications in HBM, they are not independent tools. When used together, they provide a more comprehensive basis for risk assessment. In a study assessing acrylamide exposure in lactating mothers from Valencia, forward and reverse dosimetry were integrated for comprehensive risk evaluation (S et al., 2022).

Table 1

Food-associated chemicals, corresponding biomarkers and matrices, main food sources, and human biomonitoring health-based guidance values.

Chemical	Biomarkers	Main food contributors	Matrix	Population	Exposure guidance value	HBM health-based guidance value	Ref.
Dioxins, furans, and dioxin-like PCBs	Sum of dioxins, furans, and dioxin-like PCBs	Fatty meats, dairy products, seafood	Serum	General population	PTMI, JECFA 70 pg/kg/m	BE 40-70 pg TEQ/g lipid	Aylward et al. (2008)
PCBs	Sum of PCB 138, PCB 153, and PCB 180, multiplied by two	Seafood, fatty meat, dairy products	Serum	Infants, children and women of child-bearing age	/	HBM-I 3.5 µg/L	Apel et al. (2017)
					/	HBM-II 7 µg/L	
Triclosan	Triclosan	Processed foods, packaged foods, ready-to-eat meals	Urine	6 and older	ADI, ECCC and Health Canada 0.08 mg/kg/d	BE 1783 µg/L	Faure et al. (2020a)
			Urine	Children	/	HBM-I 2000 µg/L	Apel et al. (2017)
			Urine	Adults	/	HBM-I 3000 µg/L	Apel et al. (2017)
Bisphenols							
Bisphenol A	Total Bisphenol A	Canned food, bottled drinks, packaged food	Urine	6 and older	TDI, EFSA 50 µg/kg/d	BE 2000 µg/L	Krishnan et al. (2010)
			Urine	Children	/	HBM-I 100 µg/L	Apel et al. (2017)
			Urine	Adults	/	HBM-I 200 µg/L	Apel et al. (2017)
			Urine	Children	t-TDI, EFSA 4 µg/kg/d	HBM-GV 135 µg/L	Ougier et al. (2021a)
			Urine	Adults	t-TDI, EFSA 4 µg/kg/d	HBM-GV 230 µg/L	Ougier et al. (2021a)
Bisphenol S	Total Bisphenol S	Canned food, bottled drinks	Urine	General population	LOAEL (Kolla et al., 2018, 2019; Catanese and Vandenberg, 2017) 2 µg/kg/d	HBM-GV 1 µg/L	Apel et al. (2023)
Metal/metalloid elements							
Aluminium	Aluminium	Processed foods, bakery products, grain-based foods, tea	Urine	General population	PTWI, JECFA 2 mg/kg/w	BE 123 µg/L	Poddalgoda et al. (2021b)
			Blood	General population	PTWI, JECFA 2 mg/kg/w	BE 16 µg/L	Poddalgoda et al. (2021b)
Arsenic	Sum of inorganic arsenic, MMA, and DMA	Seafood	Urine	6 and older	RfD, US EPA 0.0003 mg/kg/d	BE 6.4 µg/L	Hays et al. (2010)
Cadmium	Cadmium	Cereals, seafood	Urine	General population	MRL, ATSDR 0.0002 mg/kg/d	BE 1.2 µg/L	Hays et al. (2008)
			Urine	Children and adolescents	/	HBM-I 0.5 µg/L	Apel et al. (2017)
			Urine	Children and adolescents	/	HBM-II 2 µg/L	Apel et al. (2017)
			Urine	Adults	/	HBM-I 1 µg/L	Apel et al. (2017)
			Urine	Adults	/	HBM-II 4 µg/L	Apel et al. (2017)
			Urine	11 to 20	TWI, EFSA 2.5 µg/kg/w	HBM-GV 0.2 µg/g creatinine	Lamkarkach et al. (2021)
Mercury	Mercury	Seafood	Blood	Children and adults	/	HBM-I 5 µg/L	Apel et al. (2017)
			Blood	Children and adults	/	HBM-II 15 µg/L	Apel et al. (2017)
			Urine	Children and adults	/	HBM-I 7 µg/L	Apel et al. (2017)
			Urine	Children and adults	/	HBM-II 25 µg/L	Apel et al. (2017)

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Table 1 (continued)

Chemical	Biomarkers	Main food contributors	Matrix	Population	Exposure guidance value	HBM health-based guidance value	Ref.
Acrylamide							
Acrylamide	Acrylamide hemoglobin valine terminal adducts	Fried starchy foods, roasted coffee, biscuits, crispbread	Blood	General population	RfD, US EPA 0.002 mg/kg/d	BE 190 pmol/g Hb	Faure et al. (2020b)
Acrylamide	Glycidamide hemoglobin valine terminal adducts		Blood	General population	RfD, US EPA 0.002 mg/kg/d	BE 176 pmol/g Hb	Faure et al. (2020b)
Pesticides							
2,4-D	2,4-D	Cereals, vegetables, fruits	Urine	3 to 6	RfD, US EPA 0.21 mg/kg/d	BE 7000 µg/L	Aylward and Hays (2015)
			Urine	15 and older	RfD, US EPA 0.21 mg/kg/d	BE 10500 µg/L	Aylward and Hays (2015)
Chlorpyrifos	Chlorpyrifos	Cereals, vegetables, fruits	Blood	Infants	/	BGV 4200 ng/L	Arnold et al. (2015)
			Blood	Adults	/	BGV 6100 ng/L	Arnold et al. (2015)
	3,5,6-trichloro-2-pyridinol		Urine	Infants	/	BGV 520 µg/L	Arnold et al. (2015)
			Urine	Adults	/	BGV 2100 µg/L	Arnold et al. (2015)
Cyfluthrin	FPBA	Leafy vegetables, fruits	Urine	6 and older	ADI, JECFA 0.04 mg/kg/d	BE 400 µg/L	Hays et al. (2009)
			Urine	Children	ADI, EC, EFSA 0.01 mg/kg/d	HBM-GV 80 µg/L	Apel et al. (2023)
			Urine	Adults	ADI, EC, EFSA 0.01 mg/kg/d	HBM-GV 130 µg/L	Apel et al. (2023)
Deltamethrin	DBCA	Vegetables, fruits	Urine	6 and older	ADI, JMPR 0.01 mg/kg/d	BE 60 µg/L	Aylward et al. (2011)
Glyphosate	Glyphosate	Vegetables, fruits	Urine	Adults	ADI, Health Canada 0.3 mg/kg/d	BE 5400 µg/L	Hays et al. (2023)
PFAS							
PFOA	PFOA	Fish, meat, eggs, dairy	Serum	Adults	Draft RfD, US EPA 0.03 ng/kg/d	BE 0.8 µg/L	Tewfik et al. (2023)
			Plasma	General population	/	HBM-I 2 µg/L	Schümann et al. (2021)
			Plasma	General population	/	HBM-II 10 µg/L	Schümann et al. (2021)
			Plasma	Women of childbearing age	/	HBM-II 5 µg/L	Schümann et al. (2021)
PFOS	PFOS	Seafood, meat, offal, eggs	Plasma	General population	/	HBM-I 5 µg/L	Schümann et al. (2021)
			Plasma	General population	/	HBM-II 20 µg/L	Schümann et al. (2021)
			Plasma	Women of childbearing age	/	HBM-II 10 µg/L	Schümann et al. (2021)
Plasticizers							
Benzylbutyl phthalate	Monobenzyl phthalate	Fatty foods, plastic-packaged foods	Urine	6 and older	TDI, EFSA 0.5 mg/kg/d	BE 12000 µg/L	Aylward et al. (2009a)
Di-(2-ethylhexyl) phthalate	Sum of 5-cx-MEPP and 5-OH-MEHP	Fatty foods, plastic-packaged foods	Urine	6 to 13	TDI, EFSA 0.05 mg/kg/d	HBM-GV 380 µg/L	Lange et al. (2021)
			Urine	Adolescents and adults	TDI, EFSA 0.05 mg/kg/d	HBM-GV 570 µg/L	Lange et al. (2021)
	Sum of MEHP, MEHHP, and MEOHP		Urine	6 and older	TDI, EFSA 0.05 mg/kg/d	BE 660 µg/L	Aylward et al. (2009b)
Di-(2-ethylhexyl) terephthalate	5cx-MEPTP	Fatty foods, plastic-packaged foods	Urine	Children	/	HBM-I 1.8 mg/L	Apel et al. (2017)
			Urine	Adults	/	HBM-I 2.8 mg/L	Apel et al. (2017)
Diisononyl phthalate	Monoisononyl phthalate	Fatty foods, plastic-packaged foods	Urine	6 and older	TDI, EFSA 0.15 mg/kg/d	BE 89 µg/L	Hays et al. (2011)

Abbreviations: 2,4-D, 2,4-dichlorophenoxyacetic acid; 5-cx-MEPP, mono(5-carboxy-2-ethylpentyl) phthalate; 5-OH-MEHP, mono(2-ethyl-5-hydroxyhexyl) phthalate; 5cx-MEPTP, mono(2-ethyl-5-carboxypentyl) terephthalate; ADI, acceptable daily intake; ATSDR, Agency for Toxic Substances and Disease Registry; BE, biomonitoring equivalent; BGVs, biomonitoring guidance values; DBCA, 3-(2,2-dibromovinyl)-2,2-dimethylcyclopropane carboxylic acid; DMA, dimethylarsinic acid; EC, European Commission; ECCC, Environment and Climate Change Canada; EFSA, European Food Safety Authority; FPBA, 4-fluoro-3-phenoxybenzoic acid; Hb, hemoglobin; HBM, human biomonitoring; HBM-GV, human biomonitoring guidance value; HBM-I, human biomonitoring value I; HBM-II, human biomonitoring value II; JECFA, Joint FAO/WHO Expert Committee on Food Additives; JMPR, Joint FAO/WHO Meeting on Pesticide Residues; LOAEL, lowest observed adverse effect level; MEHHP, mono-(2-ethyl-5-hydroxyhexyl) phthalate; MEHP, mono-2-ethylhexyl phthalate; MEOHP, mono-(2-ethyl-5-oxohexyl) phthalate; MMA, monomethylarsonic acid; MRL, minimal risk level; PCB, polychlorinated biphenyl; PFAS, per- and polyfluoroalkyl substances; PFOA, perfluorooctanoic acid; PFOS, perfluorooctane sulfonate; PTMI, provisional tolerable monthly intake; Ref., reference; RfD, reference dose; t-TDI, temporary tolerable daily intake; TDI, tolerable daily intake; TEQ, toxicity equivalent; TWI, tolerable weekly intake; US EPA, United States Environmental Protection Agency.

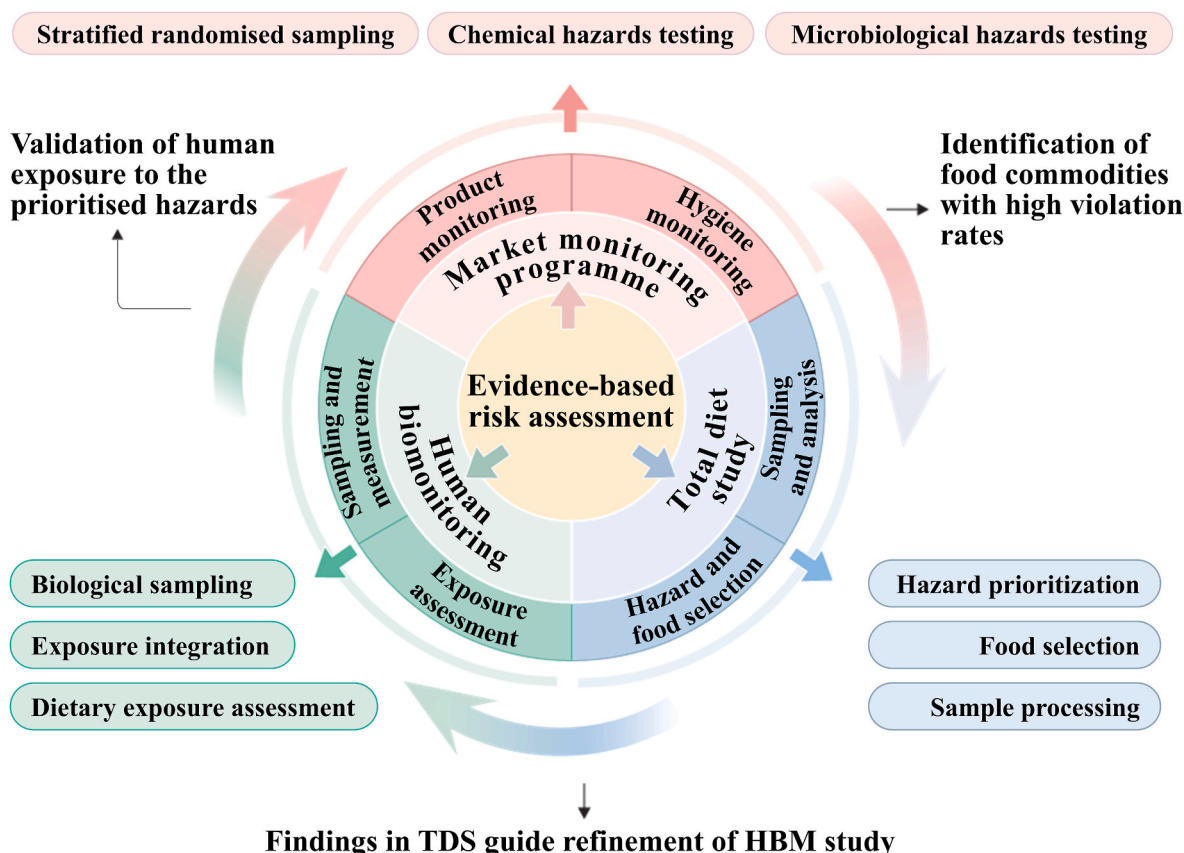


Fig. 3. A comprehensive framework for food safety risk assessment in Singapore using market monitoring, total diet study, and human biomonitoring.

5. Applications of HBM in food safety risk assessment

Although methods for interpreting HBM data are well established, the core value of HBM is realised through its practical application in risk assessment, as illustrated by the following case studies at both the national and chemical levels.

5.1. National perspectives

5.1.1. Singapore

Singapore has established an integrated food safety exposure assessment framework through the coordinated application of the National Chemical Hazard Monitoring Programme (MMP), the Total Diet Study (TDS), and HBM (Fig. 3) (Li et al., 2025a). The MMP provides baseline information on chemical hazards in retail foods through risk-based sampling, the TDS estimates population dietary exposure under realistic consumption scenarios and identifies priority hazards, and HBM captures actual internal exposure by measuring chemicals in biological samples. Together, these components form a dynamic system in which signals from food monitoring and dietary studies can be corroborated by human exposure data.

PFAS serve as a representative example of how this framework operates in practice. Monitoring under the MMP detected low levels of PFAS in seafood, including anchovies, fin-fishes, with concentrations generally lower than those reported in other countries and unlikely to pose an immediate safety concern. In contrast, findings from Singapore's TDS identified PFAS as high-priority hazards, particularly for population groups with high consumption of fish and seafood, while biomonitoring studies have reported associations between dietary intake of certain food groups and increased plasma concentrations of perfluorinated alkyl acids, underscoring the need for further targeted investigations.

5.1.2. China

The CNHBM is the first national prospective cohort established in China to characterize internal chemical exposures, identify exposure profiles and temporal trends, and build a national biorepository (Cao et al., 2021). It follows the same participants across survey cycles, each lasting two years and repeated every three years, enabling longitudinal epidemiologic follow-up. Within this evolving national HBM framework, internal exposure data are increasingly combined with dietary surveys and food monitoring to support integrated exposure assessment in China. Wan et al. (2022) combined population HBM data, dietary exposure information, and machine learning algorithms to develop models for predicting acrylamide exposure levels. For mycotoxins, Zhang et al. (2020) demonstrated concordance between external dietary exposure and internal dose by combining duplicate diet sampling with urinary biomarkers to assess zearalenone exposure.

Furthermore, Li et al. (2025b) conducted a systematic assessment of acrylamide exposure in a general population aged 45–75 years from Zhejiang Province, China. This involved integrating dietary intake estimation with internal exposure characterisation based on urinary biomarkers. Daily dietary acrylamide intake was estimated using food frequency questionnaires combined with reported acrylamide concentrations in foodstuffs. Urine samples were simultaneously collected to quantify acrylamide metabolites. Reverse dosimetry was then used to convert urinary metabolite levels into estimated daily intake and to calculate margins of exposure (MOE), with a PBTK model applied to cross-validate the intake estimates. In parallel, forward dosimetry was conducted by comparing urinary biomarkers with the BE to prioritise exposure-related risk.

Looking ahead, the integrated HBM and dietary survey approach holds significant potential for comparing internal chemical exposures across population subgroups with distinct dietary patterns, such as vegetarians and non-vegetarians or groups with high and low fish

consumption.

5.2. From single-substance to mixture risk assessment

5.2.1. Chlorpyrifos

Chlorpyrifos, an organophosphate insecticide, has undergone progressive reductions in its acceptable daily intake following evaluations by the EFSA and was banned in 2020. In this regulatory context, Tarazona et al. (2022) integrated European Union food monitoring data with HBM data from studies aligned with the HBM4EU initiative to refine chlorpyrifos risk assessment. Risk characterisation was performed using the MOE approach, defined as the ratio between the HBM points of departure and the corresponding internal exposure level. Internal exposure estimates were obtained either prospectively through PRIMo-based dietary modelling or retrospectively from measured HBM data. Overall, this case illustrates how HBM data can complement exposure assessment and support robust single-substance risk characterisation, particularly when health-based guidance values cannot be established.

5.2.2. Mixture risk assessment in the PARC framework

People are exposed to a wide range of chemicals through multiple sources, and there is growing consensus that the combined effects of chemical mixtures need to be addressed. In response, the Real-life mixtures project within PARC develops practical tools and methodological approaches for mixture risk assessment based on HBM data (Arapaki et al., 2024). Based on PARC priorities, four chemical families, namely pesticides, metals, PFAS and mycotoxins, were prioritised and linked to specific effects of concern, resulting in six case studies that are being addressed within the Real-life mixtures project and are summarised in Table 2.

Among these, pesticides with effects on brain function and/or acetylcholinesterase inhibition serve as a representative case study illustrating the application of the PARC mixture risk assessment framework. Reverse dosimetry was applied to convert urinary

metabolite concentrations into estimated daily intakes of parent compounds. For each pesticide, MOEs were calculated by comparing estimated daily intakes with no observed adverse effect levels (NOAELs) derived from EFSA, and combined exposure was characterised using the total MOE.

6. Limitations and recommendations for the integration of HBM into food safety risk assessment

HBM involves analysing biomarkers in biological samples to provide a more reliable measurement of the internal dose. This method reflects exposure from various pathways, including ingestion, inhalation, and dermal contact, and considers multiple sources, such as consumer products, personal care products, dietary intake. This reduces the need to model each source separately. Although HBM represents a valuable tool for assessing food safety risks, its practical application is limited by technical, data-related and policy constraints.

6.1. Limitations

One key limitation from a standardisation perspective is the lack of harmonised biomarkers for many chemicals, which substantially limits the comparability and integrability of HBM data across studies (Zare Jeddi et al., 2021). Although several international guidelines have been developed, including the Minimum Information Requirements for Human Biomonitoring (MIR-HBM) and the HBM Global Registry Framework, considerable heterogeneity in study design, terminology, and implementation practices persists across HBM studies. In addition, at the analytical level, validated high-sensitivity methods are not universally available, and challenges related to background contamination and quality control remain critical, particularly at low exposure levels (Yang et al., 2025). Aluminium, for example, is highly susceptible to exogenous contamination during sample collection, storage, and analysis (Ščančar and Milačič, 2006).

Beyond analytical considerations, insufficient integration of

Table 2

Summary of case studies on mixture risk assessment based on human biomonitoring data in PARC.

Chemical group	Health effect of concern	Prioritised mixture	HBM data source	Risk assessment approach	Dosimetry strategy
Pesticides	Effects on nervous system, brain and/or AChE inhibition	Chlorpyrifos, Triazophos, Omethoate, Dichlorvos, Formetanate, Carbofuran, Azinphos-ethyl, Dimethoate, Pirimiphos-methyl, Methomyl	ESTEBAN study, CELSPAC-SPECIMEn cohort	Component-based mixture risk assessment	Reverse dosimetry to estimated daily intakes and calculation of MOE and MOET based on NOAELs
Pesticides	Effects on nervous system, motor division	Triazophos, Deltamethrin, Omethoate, Chlormequat, Thiram, Beta-cypermethrin, Alpha-cypermethrin, Lambda-cyhalothrin, Azinphos-ethyl, Formetanate	ESTEBAN study, CELSPAC-SPECIMEn cohort	Component-based mixture risk assessment	Reverse dosimetry to estimated daily intakes and calculation of MOE and MOET based on NOAELs
Metals	Effects on kidney	Arsenic, Cadmium, Mercury, Lead	European HBM datasets	Component-based mixture risk assessment	Calculation of ratios between HBM-EL and HBM-TV for each metal, followed by summation of ratios to derive the mRPI
PFAS	Effects on Immune system	PFBA, PFHxA, PFHxS, PFNA, PFOA, PFOS	ESTEBAN study	Component-based mixture risk assessment	Conversion of individual PFAS concentrations into PFOA-equivalent internal concentrations, followed by summation of equivalents and comparison with the HBM guidance value
Mycotoxins	Effects on hematopoietic system	Aflatoxins, Alternaria toxins, Deoxynivalenol, Enniatins, Fumonisin, Nivalenol, Ochratoxin A, T2/HT2 toxin, Zearalenone	Relevant HBM studies identified within the PARC framework	To be addressed in a subsequent stage of the project	To be addressed in a subsequent stage of the project
All chemical groups	Developmental neurotoxicity	Lead, Methyl Mercury, Inorganic Arsenic	HBM studies measuring lead, methyl mercury and inorganic arsenic	Component-based mixture risk assessment	Combination of reverse dosimetry and forward dosimetry approaches

Abbreviations: HBM, human biomonitoring; MOE, margin of exposure; MOET, total margin of exposure; NOAELs, no observed adverse effect levels; HBM-EL, human biomonitoring exposure level; HBM-TV, human biomonitoring toxicological value; mRPI, modified reference point index; PARC, European Partnership for the Assessment of Risks from Chemicals; PFAS, per- and polyfluoroalkyl substances; PFBA, perfluorobutanoic acid; PFHxA, perfluorohexanoic acid; PFHxS, perfluorohexane sulfonic acid; PFNA, perfluorononanoic acid; PFOA, perfluorooctanoic acid; PFOS, perfluorooctane sulfonic acid.

toxicokinetic and toxicodynamic information complicates the selection of biological matrices and sampling time points (de Paula Nunes et al., 2026). For BPA, limited consideration of its rapid elimination, with a biological half-life of approximately 6 h, contributes to pronounced within- and between-day variability in urinary concentrations, such that single spot, first-morning, or 24-h urine samples may not reliably represent temporal exposure patterns (Ye et al., 2011). Additionally, different chemical forms can exhibit distinct metabolic and excretory pathways, as illustrated by the contrast between antimony(III) and antimony(V) (Filella et al., 2002a, 2002b). Antimony(III) is more chemically reactive and toxic, with stronger binding to thiol-containing biomolecules and slower elimination, whereas antimony(V) is more stable, more soluble, and rapidly excreted via urine. Moreover, although HBM captures internal dose, most studies continue to assess individual substances in isolation, despite the fact that real-world environmental and dietary exposures typically involve mixtures of chemicals with shared sources or toxicological pathways and the potential for combined effects, such as PFAS and pesticides (Pletz et al., 2020).

Finally, study design and implementation factors further limit the capacity of HBM to characterize exposure patterns over time. The predominance of cross-sectional designs and single time-point measurements restricts the assessment of chronic, cumulative, and life-course exposures, especially for persistent and bioaccumulative chemicals. In addition, the relatively high costs associated with participant recruitment, sample collection, and advanced analytical measurements constrain study scale and limit the feasibility of repeated or longitudinal sampling. Large-scale HBM studies also involve the collection of personal information and therefore require careful consideration of ethical and data protection issues. Ensuring that the collection, processing, and secondary use of HBM data comply with applicable privacy regulations, such as the EU General Data Protection Regulation (GDPR), is essential for safeguarding individual rights and maintaining public trust (Mondschein et al., 2019).

6.2. Recommendations

In summary, while HBM provides valuable information on chemical exposures, its current limitations restrict its application to broader risk assessments. The following recommendations aim to overcome these challenges.

6.2.1. Analytical advances in next-generation human biomonitoring

In response to emerging contaminants and increasingly complex exposure profiles, advances in analytical methodologies, together with the establishment of corresponding quality control and reference frameworks, are widely regarded as a key direction for the future of HBM. Next-generation liquid chromatography–tandem mass spectrometry platforms can now simultaneously quantify dozens to over a hundred exogenous chemicals and their metabolites, with detection limits typically in the picogram to nanogram per millilitre range (Jobst et al., 2025). Furthermore, longitudinal breast milk analyses have revealed pyrrolizidine and tropane alkaloids in this matrix for the first time, highlighting the ability of such platforms to characterize early-life exposure factors (Jamnik et al., 2022).

In recent years, non-targeted analysis (NTA) has attracted increasing attention because of its flexibility. For example, Oesterle et al. (2023) developed an NTA and suspect screening workflow using high-resolution mass spectrometry to analyze compounds in human biological fluids. High-resolution mass spectrometry enables the comprehensive annotation of chemical features in real samples, with annotation further aided by online resources such as PhytoHub and Exposome-Explorer. Given current limitations in definitive structural confirmation and quantitative accuracy, further research integrating high-resolution mass spectrometry with low-resolution mass spectrometry is recommended to strengthen validation and downstream interpretation.

6.2.2. Integrating human biomonitoring with network analysis for mixture risk assessment

Recent methodological advances demonstrate that integrating HBM data with network analysis provides a powerful, data-driven approach to characterising real-life chemical mixtures. Proof-of-concept studies have shown that correlation- and dependency-based networks constructed from HBM measurements can identify coherent communities of co-occurring biomarkers, reflecting common exposure sources, shared physicochemical properties, or similar toxicokinetic behaviour, thereby moving beyond single-chemical assessments toward mixture-oriented exposure characterisation (Rodriguez Martin et al., 2023). Network-based approaches applied to cord blood and population-based HBM datasets further illustrate that these exposure communities are not static but can vary across time, demographic strata, and key covariates such as maternal smoking status or body mass index, highlighting their relevance for life-course and susceptible-population analyses (Ottenbros et al., 2021). Importantly, by identifying sets of chemicals that consistently cluster together in real populations, network analysis offers a rational basis for prioritising mixtures for downstream hazard characterisation. When combined with aggregate exposure pathway and adverse outcome pathway frameworks, these empirically derived exposure communities can be mapped onto shared biological targets or key events, thereby strengthening chemical mixture risk assessment by linking internal exposure patterns with mechanistic understanding and potential health outcomes.

6.2.3. Long-term human biomonitoring projects and data management

Future developments in HBM are expected to place greater emphasis on long-term exposure assessment to achieve a more comprehensive understanding of exposure patterns. In addition, the establishment of global or regional HBM data standards is essential to enable reliable data sharing and international comparisons.

7. Conclusion

Food safety is fundamental to global public health and sustainable development. HBM has emerged as a valuable tool that complements and validates traditional external exposure assessments by providing direct and reliable evidence to characterize exposure patterns in specific populations and identify potential health risks. Despite these advantages, the broader application of HBM faces methodological challenges such as limited standardisation and complexities in data interpretation, as well as practical constraints including high costs and ethical or privacy considerations. Therefore, it is crucial to advance analytical methods in next-generation HBM and integrate it with complementary approaches. It is also important to foster long-term HBM projects alongside effective data management. With the continued refinement of methods for interpreting data, HBM is expected to play a central role in food safety risk assessment, providing robust scientific support for public health decision-making.

CRediT authorship contribution statement

Caiting Bu: Writing – original draft, Writing – review & editing, Visualization, Investigation, Formal analysis, Data curation. **Daoyuan Yang:** Writing – review & editing, Visualization, Formal analysis. **Haixia Sui:** Writing – original draft, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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Abbreviations

BEs, biomonitoring equivalents; BGVs, biomonitoring guidance values; BPA, bisphenol A; CHMS, Canadian Health Measures Survey; CNHBM, China National Human Biomonitoring Programme; ECF, exposure conversion factor; EFSA, European Food Safety Authority; GDPR, General Data Protection Regulation; HBM, human biomonitoring; HBM-GVs, human biomonitoring guidance values; HBM-I, Human Biomonitoring I value of the German Human Biomonitoring Commission; HBM-II, Human Biomonitoring II value of the German Human Biomonitoring Commission; HBM4EU, Human Biomonitoring for Europe; HTTK, High-Throughput Toxicokinetics; ICF, IndusChemFate; MCMC, Markov chain Monte Carlo; MIR-HBM, minimum information requirements for human biomonitoring; MMP, Monitoring Programme; MOE, margin of exposure; NHANES, National Health and Nutrition Examination Survey; NOAELs, no observed adverse effect levels; NTA, non-targeted analysis; PARC, Partnership for the Assessment of Risks from Chemicals; PBPK, physiologically based pharmacokinetic; PBTK, physiologically based toxicokinetic; PFAS, per- and polyfluoroalkyl substances; PFHxS, perfluorohexane sulfonic acid; PFNA, perfluorononanoic acid; PFOA, perfluorooctanoic acid; PFOS, perfluorooctane sulfonic acid; TDS, Total Diet Study.

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